QUANTITATIVE RESEARCH / OCTOBER 2026

When does aggressive quoting pay?

Inventory risk, informed flow, and the cost of stale prices

An event-driven synthetic study prepared for Thomas Krane. 12,580 sessions across 595 exogenous market tapes; four policy ablations.

The useful distinction is between reducing inventory exposure and selecting better trades. Inventory skew substantially reduced risk, but at tight widths it did not remove toxic execution losses. Volatility widening changed the filled-order mix and the edge earned per trade. A tuned static baseline explains part of the apparent profit advantage of the combined policy.

Reference experiment: mean net P&L

Regime	Fixed	Inventory	Volatility	Combined
Benign	$111.70	$112.70	$132.51	$134.68
Toxic	-$22.79	-$27.28	$26.66	$30.17
Vol. burst	$81.02	$82.14	$120.91	$111.47
Jump shock	$80.73	$71.11	$103.78	$92.29
Thin	$2.80	$6.38	$19.33	$16.34
High costs	$51.13	$51.36	$81.47	$83.03

40 held-out seeds per regime. Base half-spread $0.04 for each policy; actual widths differ when the volatility feature is active. 30-minute sessions, one-share passive fills, $100 starting price, 30-share cap. Paid close is included.

Three conclusions to defend

1. Aggressiveness is conditional. Tightening from an overly wide quote can increase ordinary-flow profit. Tightening further sacrifices edge and increases the share of toxic fills; fees can turn high turnover into a loss.

2. Risk control is a separate result. In toxic flow, inventory-only quoting reduced average absolute inventory from 15.23 to 1.44 shares, while mean net P&L remained -$27.28.

3. Adaptation has limits. Development-selected fixed spreads narrowed the profit gap. Slower refresh and unanticipated shocks remain failure cases. These results measure behavior within the specified simulation, not a validated trading edge.

The delivered artifacts include source code, all run-level metrics, illustrative fill ledgers, a local dashboard, an offline explorer, and an interview guide.

A causal market model

All monetary quantities are dollars per share. The delivered study uses one-second steps.

Information and execution order

Within step t	What happens
1 / Observe	The policy sees public mid[t], its own inventory, and prior returns. EWMA variance uses only price changes already observed.
2 / Quote	At scheduled refreshes, bid/ask prices are rounded outward to ticks. Quotes are constrained to remain passive relative to the observed midpoint.
3 / Move and fill	The efficient price moves to mid[t+1]. Ordinary flow uses mid[t] plus random depth; informed flow sees the new value. Independently ordered one-share events test the standing quotes.
4 / Mark and close	Inventory is capped at each fill. Wealth is marked at the efficient price. At the session end, residual shares are liquidated with spread, slippage, and fees.

Quotes and feature ablations

```
Center r = observed mid - gamma * inventory * EWMA variance * horizon.
Width h = base half-spread + 0.5 * gamma * variance * horizon
+ volatility buffer * sqrt(variance * refresh exposure).
```

Policy	Center shifts with inventory	Width changes with variance
Fixed	No	No
Inventory	Yes	No
Volatility	No	Yes
Combined	Yes	Yes

Gamma = 0.12; risk horizon = min(40 seconds, time remaining); EWMA alpha = 0.04; common initial sigma = 0.025; volatility buffer = 0.8; tick = $0.01. The inventory center is motivated by Avellaneda-Stoikov. This implementation adds causal variance estimates and an explicit buffer; it is a heuristic rather than the paper's optimal-control solution.

Market assumptions

The efficient price is an arithmetic diffusion plus compound-Poisson normal jumps. Six regimes vary volatility, jump size, ordinary/informed arrival rates, and fees. The burst regime changes volatility in the middle third; the shock regime includes a random-sign $1.50 jump at the midpoint. Ordinary willingness-to-pay depth is exponential with mean $0.10. Informed trades require at least $0.005 edge.

An immutable tape is created before policy evaluation from separate random streams for prices, ordinary flow, informed flow, and arrival priority. Strategy actions never consume random numbers or influence later market events.

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Every dollar has an explanation

Independent cash/inventory updates must reconcile with the component sum.

Let dq be the dealer's signed one-share fill, p its price, m_ref the midpoint used to form the quote, and m_exec the efficient price at execution. The step begins with inventory q_before.

```
Quote edge = dq * (m_ref - p)
Fill repricing = dq * (m_exec - m_ref)
Inventory P&L = q_before * (m_exec - m_previous)
Net P&L = sum of those components - fees - paid close
```

Quote edge includes inventory skew and tick rounding. Fill repricing can be positive or negative. Summing quote edge and repricing gives execution edge; markouts are diagnostics and are never deducted again.

A stale-ask example

Quote reference $100; ask $100.04; new efficient price $100.20. An informed buyer lifts the ask, so the dealer sells one share. Quote edge is +$0.04 and repricing is -$0.20: execution P&L is -$0.16 before fees and closing costs. Selling above an old midpoint does not imply a profitable trade.

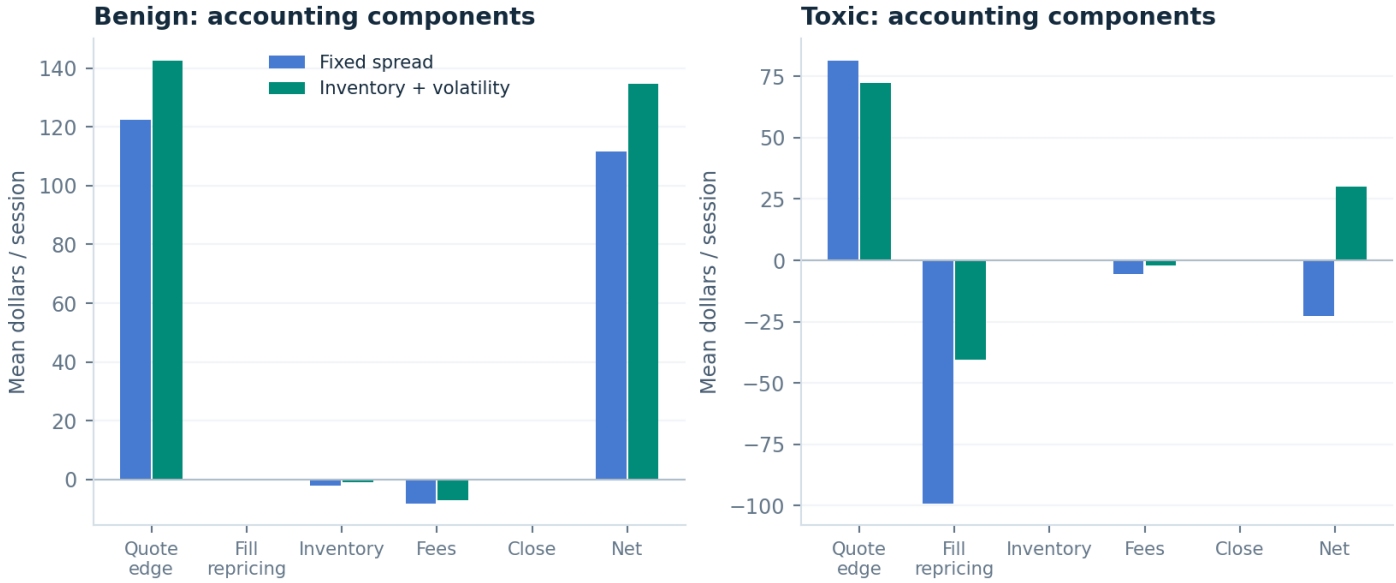


Figure 1. Mean component sums at base half-spread $0.04. Costs are shown as negative contributions; net includes paid close.

Execution quality and verification

Ledgers record trader type, signed quantity, quote reference, execution midpoint, inventory after fill, fee, and 1/5/20-second markouts. Horizons extending past session close are excluded. Informed execution losses are negative by the oracle trader definition; their size and frequency are the useful stress diagnostics.

Maximum reconciliation residual across all 12,580 sessions: 5.22e-11 dollars. The 33 tests include hand-computed fills, paid liquidation, exact fee effects, per-fill caps, future-price prefix invariance, shared tapes, and a statistical exponential fill-rate check.

QUOTELAB / RESEARCH NOTE

Aggressiveness has a frontier

Confidence bands describe uncertainty across held-out synthetic sessions.

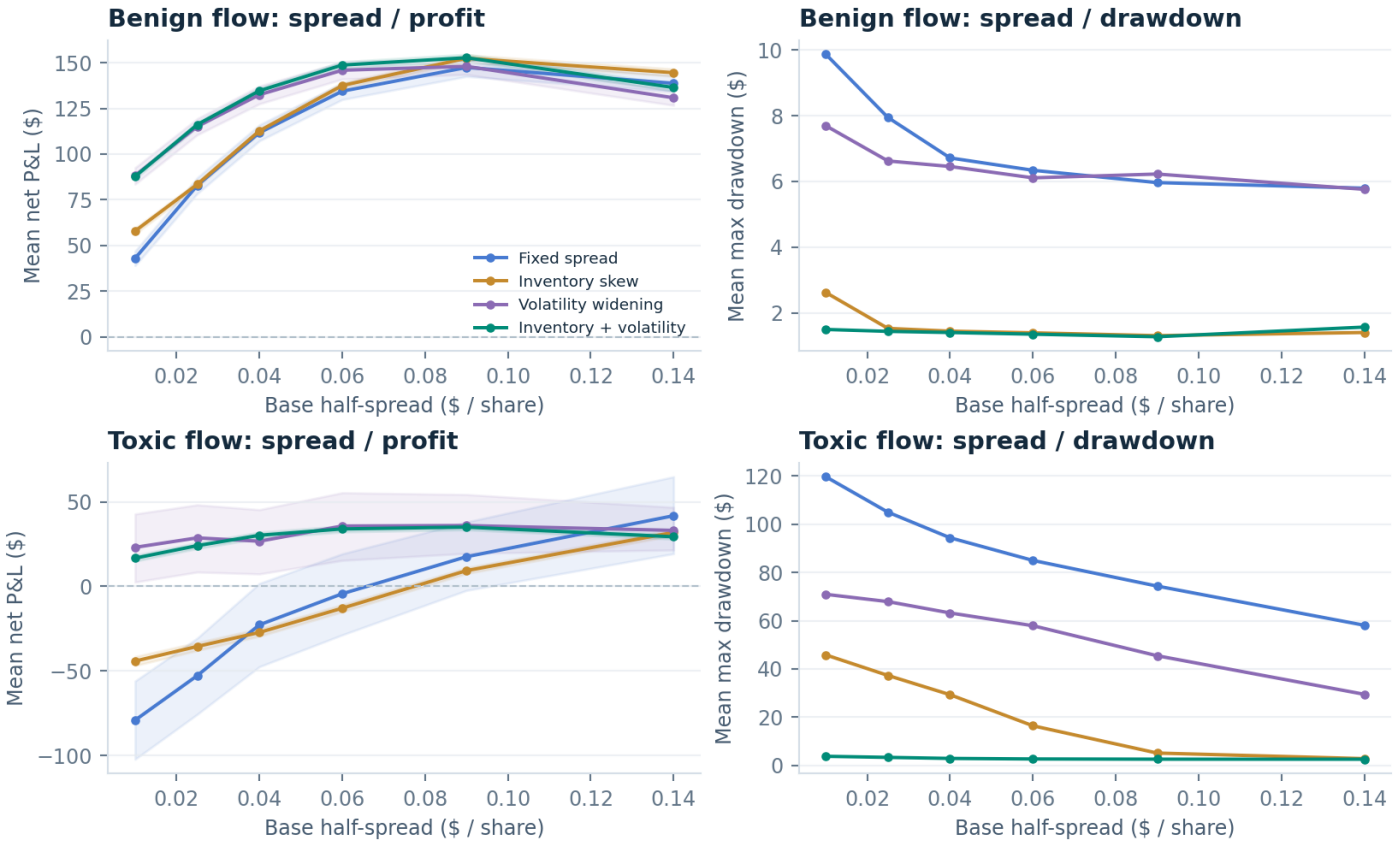


Figure 2. Tight quotes can increase turnover while decreasing P&L. Inventory skew and volatility widening have different effects on drawdown.

Illustrative paired width contrasts

Regime	Tighten base width	P&L change	Paired 95% interval
Benign	$0.14 to $0.09	$8.64	[5.21, 11.94]
Toxic	$0.09 to $0.04	-$40.30	[-58.48, -21.56]
High costs	$0.04 to $0.01	-$89.63	[-92.07, -86.86]

These width contrasts are exploratory examples from the fixed grid, not a claim that one optimum transfers between regimes. All use the same seeds within each contrast.

A simple expected-value explanation

With only ordinary exponential-depth flow, fresh symmetric quotes, no caps, and no inventory risk, expected profit rate is proportional to (distance - fee) * exp(-k * distance). Its stationary maximum is at distance = fee + 1/k. Tightening helps when moving toward that maximum from above. Informed execution losses and stale-reference exposure alter the calculation, so maximizing fill count is an insufficient objective.

A fairer baseline changes the claim

One width per policy is selected on development seeds and then frozen.

Development uses 30 seeds per regime; evaluation uses 40 disjoint seeds. Selection assigns equal weight to all six regimes and maximizes mean P&L - 0.25 * mean maximum drawdown. The score weight is a stated research choice, not an economic optimum.

Frozen base widths: Inventory + volatility $0.090; Fixed spread $0.090; Inventory skew $0.140; Volatility widening $0.090.

Regime	Fixed	Inventory	Volatility	Combined
Benign	$147.43	$144.56	$148.04	$152.76
Toxic	$17.51	$31.86	$35.95	$35.03
Vol. burst	$127.83	$114.28	$119.22	$114.69
Jump shock	$102.32	$93.30	$95.94	$94.95
Thin	$18.03	$17.10	$18.75	$17.15
High costs	$110.46	$121.77	$116.83	$121.28

Selected-width held-out mean P&L. Selected widths can differ across policies; the inventory parameters and model assumptions remain unchanged.

What the feature ablations support

At the reference width, volatility-only quoting explains much of the profit improvement but retains substantially more inventory risk than the combined policy. Inventory-only quoting removes much of that risk without reliably increasing mean profit. After width selection, the combined policy has lower mean P&L than fixed quoting in some regimes; an adaptive policy is not a universal profit improvement.

Regime	Combined - fixed	Paired 95% interval	Mean drawdown: fixed / combined
Benign	$5.33	[0.93, 9.97]	5.96 / 1.27
Toxic	$17.52	[-2.54, 37.34]	74.34 / 2.50
Vol. burst	-$13.14	[-26.90, 1.44]	35.27 / 1.83

Intervals and inference

Each interval resamples entire seed-level sessions 3,000 times. For policy effects, paired differences are computed first. Inference does not count correlated fills as independent trials. Intervals measure Monte Carlo uncertainty within this model; exploratory comparisons have no multiplicity correction. Forty evaluation seeds are insufficient for precise rare-tail estimates.

QUOTELAB / RESEARCH NOTE

Failure cases and the next test

A model earns trust by making its assumptions and failures easy to inspect.

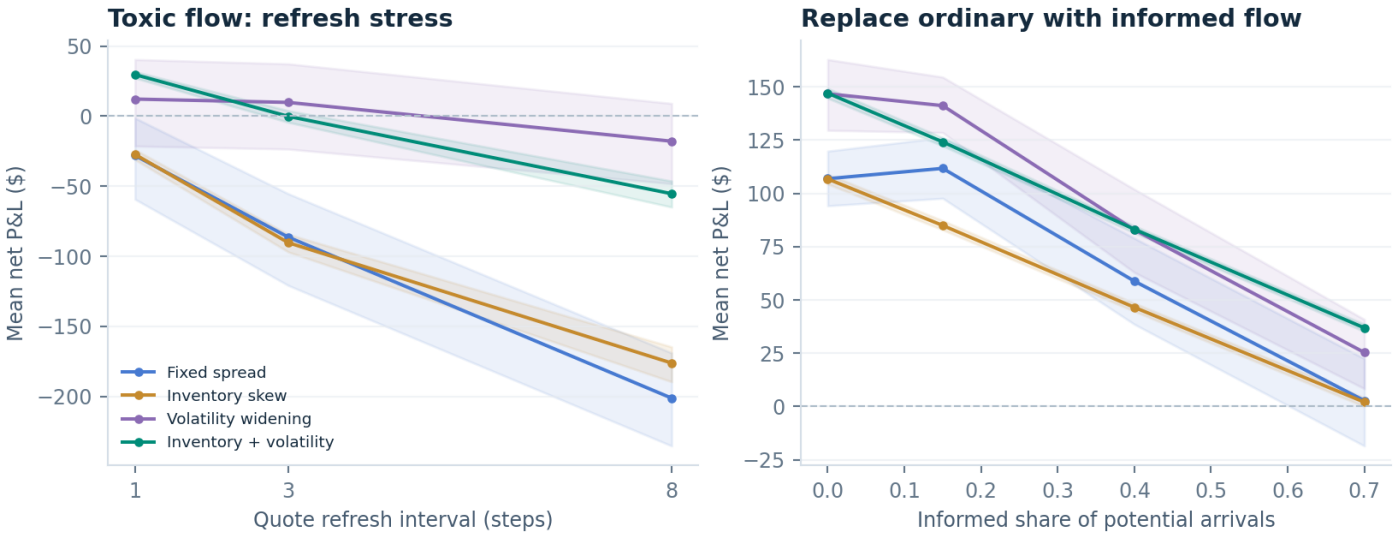


Figure 3. Additional sensitivity seeds: 25 per setting. Left: toxic-flow refresh stress. Right: informed share changes at a fixed total potential order rate of 2.4/second, base half-spread $0.04.

A measurable failure

In toxic flow, the combined policy's mean P&L changes from $29.67 at one-step refresh to -$55.37 at eight-step refresh. Its variance estimate reacts after a jump; it cannot preempt a fresh price innovation. Quotes and inventory correction both become stale between refreshes.

Limitations that matter

Instant refills, independent ordinary depth, and oracle informed traders simplify execution. There are no competitors, queues, impact, funding requirements, or network cancel dynamics. Seed holdout checks simulation selection; it does not establish market generalization. Forty evaluation paths do not identify rare-tail risk.

The next experiment

Add finite displayed size, price-time-priority queues, and delayed cancellation or replenishment. Predict the direction of the fill-mix change before running. Then test whether the conclusion survives with fresh development/evaluation seeds. Real-data replay should use conservative queue assumptions and a chronological calibration/validation split.

Reproduce / inspect

```
python -m quotelab research --workers 4
python scripts/build_report.py
python -m quotelab dashboard
```

Audit metrics, ledgers, paths, and seeds in results/. Open the offline explorer in a browser; use the local app to rerun settings.

Primary references:
Avellaneda & Stoikov (2008). High-frequency trading in a limit order book.
Jane Street. Probability & Markets guide.